

DQuez Carr
Module 2: Lab Exercise
Intelligence Gathering

Part A: Setting up VM IP Addresses

Networking Configuration on the META VM

- Cropped screenshot of information confirmation output before you enter “y”

```
msfadmin@metasploitable:/$ cd /home/msfadmin/
msfadmin@metasploitable:~$ ls
change-ip.sh  image.jpg  index.html  t  vulnerable
msfadmin@metasploitable:~$

msfadmin@metasploitable:~$ sudo sh ./change-ip.sh 192.168.1.3 192.168.1.1 255.25
5.255.0 192.168.1.255
[sudo] password for msfadmin:
*****
Changing network interface configuration
*****
address 192.168.1.3
gateway 192.168.1.1
netmask 255.255.255.0
broadcast 192.168.1.255

Verify information before proceeding.
Are you sure? (y/n):
```

```
Overwriting network configuration...
* Reconfiguring network interfaces... [ OK ]
Updated network configuration ...
eth0      Link encap:Ethernet  HWaddr bc:24:11:45:6b:0e
          inet addr:192.168.1.3  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::be24:11ff:fe45:6b0e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:39 errors:0 dropped:0 overruns:0 frame:0
          TX packets:90 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3343 (3.2 KB)  TX bytes:0 (0.0 B)

msfadmin@metasploitable:~$
```

- Cropped screenshot of ifconfig and ping commands.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr bc:24:11:45:6b:0e
          inet addr:192.168.1.3  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::be24:11ff:fe45:6b0e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:60 errors:0 dropped:0 overruns:0 frame:0
          TX packets:132 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5011 (4.8 KB)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:143 errors:0 dropped:0 overruns:0 frame:0
          TX packets:143 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:44197 (43.1 KB)  TX bytes:44197 (43.1 KB)

msfadmin@metasploitable:~$ _
```

```

msfadmin@metasploitable:~$ ping 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
  bytes from 1.1.1.1: icmp_seq=1 ttl=47 time=19.2 ms
  bytes from 1.1.1.1: icmp_seq=2 ttl=47 time=14.5 ms
  bytes from 1.1.1.1: icmp_seq=3 ttl=47 time=14.3 ms
  bytes from 1.1.1.1: icmp_seq=4 ttl=47 time=14.3 ms
  bytes from 1.1.1.1: icmp_seq=5 ttl=47 time=14.4 ms
  bytes from 1.1.1.1: icmp_seq=6 ttl=47 time=14.5 ms
  bytes from 1.1.1.1: icmp_seq=7 ttl=47 time=14.4 ms
  bytes from 1.1.1.1: icmp_seq=8 ttl=47 time=14.4 ms
  bytes from 1.1.1.1: icmp_seq=9 ttl=47 time=14.4 ms
  bytes from 1.1.1.1: icmp_seq=10 ttl=47 time=14.3 ms
  bytes from 1.1.1.1: icmp_seq=11 ttl=47 time=14.4 ms
  bytes from 1.1.1.1: icmp_seq=12 ttl=47 time=14.4 ms

--- 1.1.1.1 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 10994ms
rtt min/avg/max/mdev = 14.366/14.845/19.294/1.348 ms
msfadmin@metasploitable:~$

```

Networking Configuration on the KALI VM

- Cropped screenshot of information confirmation output before you enter “y”

```

student@ncy670-kali:~$ cd /home/student/
student@ncy670-kali:~$ ls
armitage-tmp  Desktop  Downloads  nessus_rest.rb  Pictures  Templates
change-ip.sh  Documents  Music      nessus_rest.rb.bkup  Public    Videos
student@ncy670-kali:~$ sudo sh ./change-ip.sh 192.168.1.4 192.168.1.1 255.255.255.0 192.168.1.255
[sudo] password for student:
*****
Changing network interface configuration
*****
address 192.168.1.4
gateway 192.168.1.1
netmask 255.255.255.0
broadcast 192.168.1.255

Verify information before proceeding.
Are you sure? (y/n):

```

```

Verify information before proceeding.
Are you sure? (y/n): y

Overwriting network configuration...
Restarting networking (via systemctl): networking.service.
Updated network configuration ...
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1450
    inet 192.168.1.4 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::be24:11ff:fe75:eaf9 prefixlen 64 scopeid 0x20<link>
    ether bc:24:11:75:ea:f9 txqueuelen 1000 (Ethernet)
    RX packets 236 bytes 33517 (32.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 4183 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

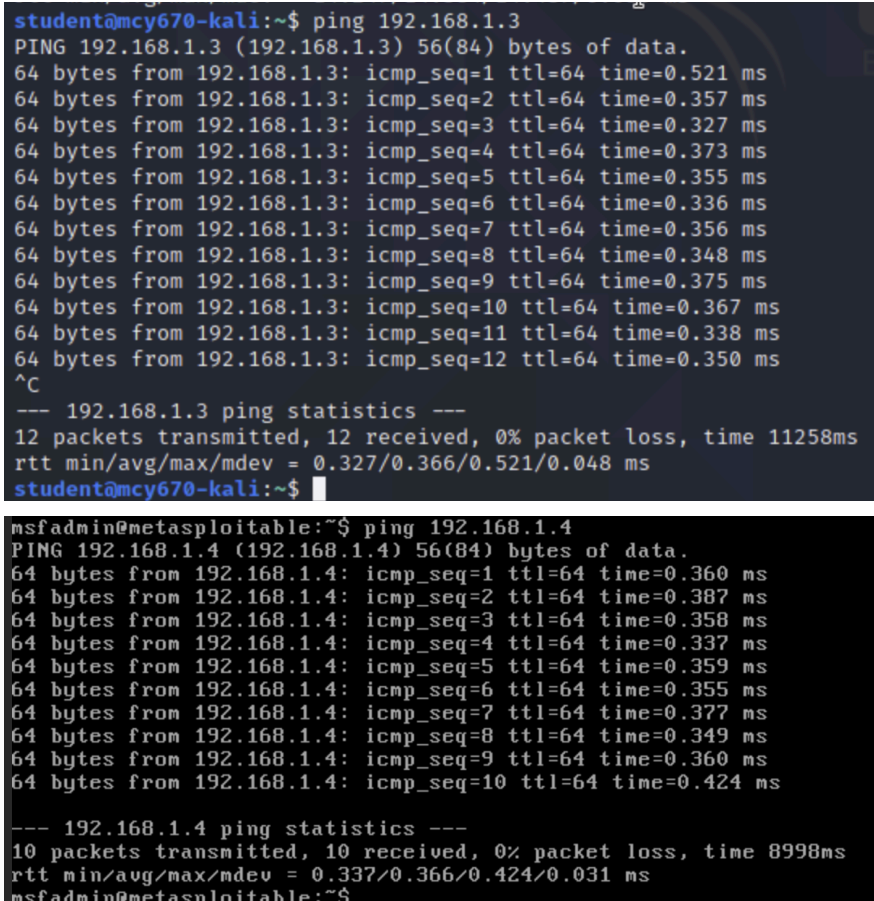
student@ncy670-kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1450
    inet 192.168.1.4 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::be24:11ff:fe75:eaf9 prefixlen 64 scopeid 0x20<link>
    ether bc:24:11:75:ea:f9 txqueuelen 1000 (Ethernet)
    RX packets 239 bytes 33791 (32.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 31 bytes 4773 (4.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 16 bytes 796 (796.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 16 bytes 796 (796.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

student@ncy670-kali:~$

```

- Cropped screenshot of ifconfig and ping commands.



```
student@mcy670-kali:~$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data.
64 bytes from 192.168.1.3: icmp_seq=1 ttl=64 time=0.521 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=64 time=0.357 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=64 time=0.327 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=64 time=0.373 ms
64 bytes from 192.168.1.3: icmp_seq=5 ttl=64 time=0.355 ms
64 bytes from 192.168.1.3: icmp_seq=6 ttl=64 time=0.336 ms
64 bytes from 192.168.1.3: icmp_seq=7 ttl=64 time=0.356 ms
64 bytes from 192.168.1.3: icmp_seq=8 ttl=64 time=0.348 ms
64 bytes from 192.168.1.3: icmp_seq=9 ttl=64 time=0.375 ms
64 bytes from 192.168.1.3: icmp_seq=10 ttl=64 time=0.367 ms
64 bytes from 192.168.1.3: icmp_seq=11 ttl=64 time=0.338 ms
64 bytes from 192.168.1.3: icmp_seq=12 ttl=64 time=0.350 ms
^C
--- 192.168.1.3 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11258ms
rtt min/avg/max/mdev = 0.327/0.366/0.521/0.048 ms
student@mcy670-kali:~$

msfadmin@metasploitable:~$ ping 192.168.1.4
PING 192.168.1.4 (192.168.1.4) 56(84) bytes of data.
64 bytes from 192.168.1.4: icmp_seq=1 ttl=64 time=0.360 ms
64 bytes from 192.168.1.4: icmp_seq=2 ttl=64 time=0.387 ms
64 bytes from 192.168.1.4: icmp_seq=3 ttl=64 time=0.358 ms
64 bytes from 192.168.1.4: icmp_seq=4 ttl=64 time=0.337 ms
64 bytes from 192.168.1.4: icmp_seq=5 ttl=64 time=0.359 ms
64 bytes from 192.168.1.4: icmp_seq=6 ttl=64 time=0.355 ms
64 bytes from 192.168.1.4: icmp_seq=7 ttl=64 time=0.377 ms
64 bytes from 192.168.1.4: icmp_seq=8 ttl=64 time=0.349 ms
64 bytes from 192.168.1.4: icmp_seq=9 ttl=64 time=0.360 ms
64 bytes from 192.168.1.4: icmp_seq=10 ttl=64 time=0.424 ms
--- 192.168.1.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 8998ms
rtt min/avg/max/mdev = 0.337/0.366/0.424/0.031 ms
msfadmin@metasploitable:~$
```

Part B: External intelligence gathering from KALI VM

Phase 1: Basic domain information gathering

- Cropped screenshot of “whois” command.

```

msf6 > whois itsecgames.com
[*] exec: whois itsecgames.com

Domain Name: ITSECGAMES.COM
Registry Domain ID: 1721761149_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-05-22T10:55:31Z
Creation Date: 2012-05-21T13:35:16Z
Registry Expiry Date: 2027-05-21T13:35:16Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS53.DOMAINCONTROL.COM
Name Server: NS54.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-10-30T00:47:57Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois

```

```

Registrars.
Domain Name: itsecgames.com
Registry Domain ID: 1721761149_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-05-22T05:55:30Z
Creation Date: 2012-05-21T08:35:16Z
Registrar Registration Expiration Date: 2027-05-21T08:35:16Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=itsecgames.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:

```

- Explanation of output on registrar, update/creation/expiration dates, name servers, and any other interesting information.

Important domain registration information is provided by the itsecgames.com WHOIS lookup results. The WHOIS server for the name is whois.godaddy.com, and it is registered through GoDaddy.com, LLC. It was last updated on May 22, 2025, and it was created on May 21, 2012. It will expire on May 21, 2027. This indicates that the domain has been in use for more than ten years and will continue to be registered for at least another two.

The domain conceals the identity of the real registrant by using Domains by Proxy, LLC in Tempe, Arizona (U.S.) for privacy protection. Name servers NS53.DOMAINCONTROL.COM and NS54.DOMAINCONTROL.COM are specified; they are typical for domains hosted by GoDaddy. The domain does not have DNS Security Extensions set, as indicated by the DNSSEC being listed as unsigned—possibly a small security lapse. All things considered, the results verify that itsecgames.com is a well-established domain with private registration, management based on GoDaddy, and normal safeguards against unwanted domain modifications.

- Cropped screenshots of Netcraft (<http://searchdns.netcraft.com>) output.



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Background

Site title	bWAPP, a buggy web application!	Date first seen	July 2012
Site rank	23976	Primary language	English
Description	Not Present		

Network

Site	http://www.itsecgames.com	Domain	itsecgames.com
Netblock Owner	XL Internet Services Amsterdam Network	Nameserver	ns53.domaincontrol.com
Hosting company	team.blue	Domain registrar	godaddy.com
Hosting country	NL	Nameserver organisation	whois.wildwestdomains.com
IPv4 address	31.3.96.40 (VirusTotal)	Organisation	Unknown
IPv4 autonomous systems	AS20857	DNS admin	dns@jomax.net
IPv6 address	Not Present	Top Level Domain	Commercial entities (.com)
IPv6 autonomous systems	Not Present	DNS Security Extensions	Enabled
Reverse DNS	web.mmebvba.com		

IP delegation

SSL/TLS

This is not a HTTPS site. If you're looking for SSL/TLS information try the [HTTPS site report](#).

Sender Policy Framework

A host's Sender Policy Framework (SPF) describes who can send mail on its behalf. This is done by publishing an SPF record containing a series of [rules](#). Each rule consists of a qualifier followed by a specification of which domains to apply this qualifier to. For more information please see open-spf.org.

Qualifier	Mechanism	Argument
+ (Pass)	mx	
+ (Pass)	a	
+ (Pass)	include	spf.protection.outlook.com
+ (Pass)	include	servers.mcsv.net
+ (Pass)	include	mme-srv-dc1.mme.local
- (Fail)	all	

Site Technology (fetched 22 days ago)

Client-Side

Includes all the main technologies that run on the browser (such as JavaScript and Adobe Flash).

Technology	Description	Popular sites using this technology
JavaScript ↗	Widely-supported programming language commonly used to power client-side dynamic content on websites	www.twitch.tv , www.netflix.com , www.msn.com

Character Encoding

A character encoding system consists of a code that pairs each character from a given repertoire with something else such as a bit pattern, sequence of natural numbers, octets, or electrical pulses in order to facilitate the transmission of data (generally numbers or text) through telecommunication networks or for data storage.

Technology	Description	Popular sites using this technology
UTF8 ↗	UCS Transformation Format 8 bit	www.deepl.com , www.tiktok.com , www.canva.com

HTTP Compression

HTTP compression is a capability that can be built into web servers and web clients to make better use of available bandwidth, and provide greater transmission speeds between both.

Technology	Description	Popular sites using this technology
------------	-------------	-------------------------------------

HTTP Compression

HTTP compression is a capability that can be built into web servers and web clients to make better use of available bandwidth, and provide greater transmission speeds between both.

Technology	Description	Popular sites using this technology
Gzip Content Encoding ↗	Gzip HTTP Compression protocol	www.amazon.in , www.amazon.es , www.jobthai.com

Doctype

A Document Type Declaration, or DOCTYPE, is an instruction that associates a particular SGML or XML document (for example, a webpage) with a Document Type Definition (DTD).

Technology	Description	Popular sites using this technology
HTML5 ↗	Latest revision of the HTML standard, the main markup language on the web	mail.google.com , x.com , webmail.vincihoteles.com

CSS Usage

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation semantics (the look and formatting) of a document written in a markup language (such as XHTML).

Technology	Description	Popular sites using this technology
CSS Media Query	No description	www.imdb.com , www.microsoft.com , www.paypal.com
External ↗	Styles defined within an external CSS file	www.linkedin.com , saas-aftral.octime.net , learn.microsoft.com

- Explanation of output on network, hosting history, and technologies used.

Itsecgames.com's Netcraft report offers comprehensive information about the site's network, hosting background, and technologies. A defective web application called bWAPP is hosted on the domain itsecgames.com, which was discovered for the first time in July 2012, indicating that it has been operational for more than ten years. GoDaddy serves as the domain registrar for the website, which is hosted by team.blue under the XL Internet Services Amsterdam Network in the Netherlands (NL). Reverse DNS is listed as web.mmebvba.com, and DNS security extensions are enabled, indicating further DNS integrity checks. The IPv4 address is 31.3.96.40.

Regarding web technologies, the website employs HTML5 for page structure, UTF-8 for character encoding, Gzip content encoding for HTTP compression (which increases load speed and efficiency), and JavaScript on the client side. For design versatility, it makes use of external

stylesheets and CSS Media Queries. But according to the research, the website does not employ HTTPS, which means SSL/TLS encryption is not there, which could pose a security risk to the confidentiality of the data. To stop email spoofing, the Sender Policy Framework (SPF) is set up properly, designating approved mail servers. Ultimately, even though the website makes use of common web technologies, its otherwise well-configured network infrastructure is seriously compromised by the absence of HTTPS.

- Responses to questions:

- How can this information help an attacker to execute successful attacks?

The report from Netcraft provides an attacker with a concise reconnaissance map:

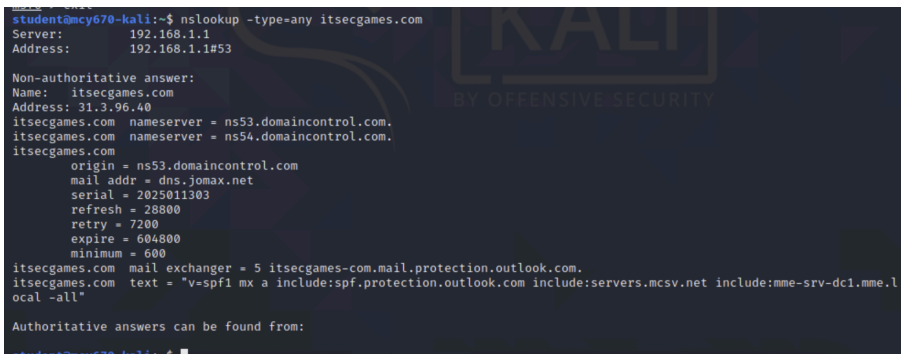
IP/ASN and hosting information indicates where to target scans and whether co-hosted sites or provider misconfigurations exist; WHOIS/registrant and creation/expiry information facilitates social-engineering or takeover opportunities; technology fingerprints (JS, UTF-8, Gzip, HTML5) and the absence of HTTPS indicate likely attack classes (XSS, unencrypted session interception, cookie tampering); and DNS/SPF/DNSSEC data direct attempts at email spoofing or DNS tampering. With that information, an attacker can focus on the simplest, most lucrative vectors (unpatched services, exposed backups, phishing paths). In response, defenders should activate TLS, lock registrar/DNS controls, eliminate exposed artifacts, patch components, and keep an eye on the listed IPs and WHOIS for any unusual changes.

- Are the name servers self-hosted or third party-hosted?

Third party-hosted

Phase 2: Advanced domain information gathering

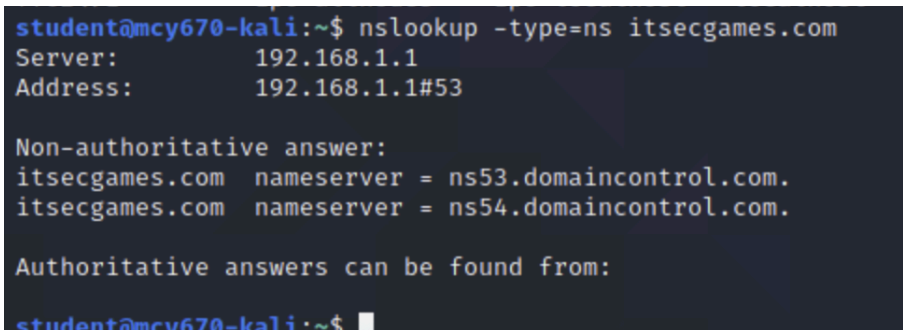
- Cropped screenshot of “nslookup” commands (variations of type parameter)



```
student@mcv670-kali:~$ nslookup -type=any itsecgames.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   itsecgames.com
Address: 31.3.96.40
itsecgames.com  nameserver = ns53.domaincontrol.com.
itsecgames.com  nameserver = ns54.domaincontrol.com.
itsecgames.com
    origin = ns53.domaincontrol.com
    mail addr = dns.jomax.net
    serial = 2025011303
    refresh = 28800
    retry = 7200
    expire = 604800
    minimum = 600
itsecgames.com  mail exchanger = 5 itsecgames-com.mail.protection.outlook.com.
itsecgames.com  text = "v=spf1 mx a include:spf.protection.outlook.com include:servers.mcsv.net include:mme-srv-dcl.mme.l
ocal -all"

Authoritative answers can be found from:
```



```
student@mcv670-kali:~$ nslookup -type=ns itsecgames.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
itsecgames.com  nameserver = ns53.domaincontrol.com.
itsecgames.com  nameserver = ns54.domaincontrol.com.

Authoritative answers can be found from:
```

```

student@mcy670-kali:~$ nslookup -type=mx itsecgames.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
itsecgames.com mail exchanger = 5 itsecgames-com.mail.protection.outlook.com.

Authoritative answers can be found from:

```

- Explanation of outputs for each command format including mail exchange and name servers. A/NS/SOA/TXT information is displayed collectively in the -type=any, the authoritative name servers are returned by -type=ns, and mail routing and priority are returned by -type=mx. Understanding the location of a domain's hosting, mail routing, and DNS zone management is made easier with the help of each returned field (A, NS, SOA, MX, and TXT).

The domain's authoritative name servers (where the zone is actually served) are listed in NS records (itsecgames.com nameserver = ns53.domaincontrol.com, ns54.domaincontrol.com). The presence of domaincontrol.com is a clear sign that the zone is being hosted by the registrar's or third-party DNS (GoDaddy/DomainControl).

The mail host(s) and a numerical preference are displayed in the MX (mail exchanger) output: mail exchanger = 5 itsecgames-com.mail.protection.outlook.com. Mail servers will try lower-value MX targets first (here mail flows via Microsoft/Outlook protection) the lower the number, the higher the priority.

- Responses to questions:
 - What is an AAAA address?
A DNS record that transfers a domain name to an IPv6 address, the more recent internet protocol intended to replace IPv4, is called a AAAA address (quad-A record). It transfers a domain to an IPv4 address similarly to an A record, except it supports the larger 128 bit IPv6 format. This guarantees that contemporary gadgets and websites can interact effectively over IPv6 networks and permits a far greater number of distinct IP addresses.
 - Would you have the jurisdiction to probe the mail servers for penetration testing?
No.

Part C: Intelligence gathering using port scanning from KALI VM

Phase 1: Using NMAP for port scanning

- META VM nmap Scanning:
 - Cropped screenshots of “nmap” commands (variations of option parameter: -sS, -Pn, -A, -v, -vv) for META VM scan.

```

root@mcy670-kali:/home/student# nmap -sS -v -Pn -oX nmapresultMeta1.txt 192.168.1.3
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times will be slower.
Starting Nmap 7.91 ( https://nmap.org ) at 2025-10-29 21:36 EDT
Initiating ARP Ping Scan at 21:36
Scanning 192.168.1.3 [1 port]
Completed ARP Ping Scan at 21:36, 0.02s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 21:36
Completed Parallel DNS resolution of 1 host. at 21:36, 0.00s elapsed
Initiating SYN Stealth Scan at 21:36
Scanning 192.168.1.3 [1000 ports]
Discovered open port 80/tcp on 192.168.1.3
Discovered open port 53/tcp on 192.168.1.3
Discovered open port 445/tcp on 192.168.1.3
Discovered open port 21/tcp on 192.168.1.3
Discovered open port 23/tcp on 192.168.1.3
Discovered open port 139/tcp on 192.168.1.3
Discovered open port 3306/tcp on 192.168.1.3
Discovered open port 5900/tcp on 192.168.1.3
Discovered open port 22/tcp on 192.168.1.3
Discovered open port 111/tcp on 192.168.1.3
Discovered open port 25/tcp on 192.168.1.3
Discovered open port 513/tcp on 192.168.1.3
Discovered open port 2121/tcp on 192.168.1.3
Discovered open port 1524/tcp on 192.168.1.3
Discovered open port 8009/tcp on 192.168.1.3
Discovered open port 1099/tcp on 192.168.1.3
Discovered open port 514/tcp on 192.168.1.3
Discovered open port 6000/tcp on 192.168.1.3
Discovered open port 6667/tcp on 192.168.1.3
Discovered open port 512/tcp on 192.168.1.3
Discovered open port 8180/tcp on 192.168.1.3

```

```

Discovered open port 5432/tcp on 192.168.1.3
Discovered open port 2049/tcp on 192.168.1.3
Completed SYN Stealth Scan at 21:36, 0.01s elapsed (1000 total ports)
Nmap scan report for 192.168.1.3
Host is up (0.00024s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: BC:24:11:45:6B:0E (Unknown)

Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (40.120KB)
root@mcy670-kali:/home/student#

```

- Explanation of outputs from the nmap scan result for META VM.
Targeting 192.168.1.3, the Nmap ran disabled host discovery (-Pn) and performed a SYN (stealth) scan (-sS). Nmap scanned 1,000 common TCP ports and discovered that the host was up. In addition to numerous other high/odd ports, it found many open ports,

including 21 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 80 HTTP, 139/445 NetBIOS/SMB, 3306 MySQL, 5432 PostgreSQL, 5900 VNC, and 6000 X11.

- Cropped screenshot of “ls” command output showing saved file

```
root@mcy670-kali:/home/student# ls
armitage-tmp  Desktop  Downloads  nessus_rest.rb  nmapresultMeta1.txt  Public  Videos
change-ip.sh  Documents  Music      nessus_rest.rb.bkup  Pictures              Templates
root@mcy670-kali:/home/student#
```

- Importing nmap scan results to msfconsole database
 - Cropped screenshots of successful imports of META VM scan results

```
msf6 > db_status
[*] Connected to msf. Connection type: postgresql.
msf6 > db_import nmapresultMeta1.txt
[*] Importing 'Nmap XML' data
[*] Import: Parsing with 'Nokogiri v1.10.10'
[*] Importing host 192.168.1.3
[*] Successfully imported /home/student/nmapresultMeta1.txt
msf6 >
```

- Cropped screenshot and short test description of output for the commands: “hosts”, “hosts -c address”, “services”

```
msf6 > hosts

Hosts
=====

```

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
192.168.1.3	bc:24:11:45:6b:0e		Unknown			device		

```
msf6 >
```

```
msf6 > hosts -c address

Hosts
=====

```

address
192.168.1.3

```
msf6 >
```

```
msf6 > services

Services
=====

```

host	port	proto	name	state	info
192.168.1.3	21	tcp	ftp	open	
192.168.1.3	22	tcp	ssh	open	
192.168.1.3	23	tcp	telnet	open	
192.168.1.3	25	tcp	smtp	open	
192.168.1.3	53	tcp	domain	open	
192.168.1.3	80	tcp	http	open	
192.168.1.3	111	tcp	rpcbind	open	
192.168.1.3	139	tcp	netbios-ssn	open	
192.168.1.3	445	tcp	microsoft-ds	open	
192.168.1.3	512	tcp	exec	open	
192.168.1.3	513	tcp	login	open	
192.168.1.3	514	tcp	shell	open	
192.168.1.3	1099	tcp	rmiregistry	open	
192.168.1.3	1524	tcp	ingreslock	open	
192.168.1.3	2049	tcp	nfs	open	
192.168.1.3	2121	tcp	ccproxy-ftp	open	
192.168.1.3	3306	tcp	mysql	open	
192.168.1.3	5432	tcp	postgresql	open	
192.168.1.3	5900	tcp	vnc	open	
192.168.1.3	6000	tcp	x11	open	
192.168.1.3	6667	tcp	irc	open	
192.168.1.3	8009	tcp	ajp13	open	
192.168.1.3	8180	tcp	unknown	open	

```
msf6 >
```


- Cropped screenshot and short test description of running nmap from msfconsole

```
msf6 > db_nmap -sS -v -Pn -oX nmapresultMeta2.txt 192.168.1.3
[*] Nmap: 'Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times will be slower.'
[*] Nmap: Starting Nmap 7.91 ( https://nmap.org ) at 2025-10-29 22:00 EDT
[*] Nmap: Initiating ARP Ping Scan at 22:00
[*] Nmap: Scanning 192.168.1.3 [1 port]
[*] Nmap: Completed ARP Ping Scan at 22:00, 0.08s elapsed (1 total hosts)
[*] Nmap: Initiating Parallel DNS resolution of 1 host. at 22:00
[*] Nmap: Completed Parallel DNS resolution of 1 host. at 22:00, 0.00s elapsed
[*] Nmap: Initiating SYN Stealth Scan at 22:00
[*] Nmap: Scanning 192.168.1.3 [1000 ports]
[*] Nmap: Discovered open port 111/tcp on 192.168.1.3
[*] Nmap: Discovered open port 5900/tcp on 192.168.1.3
[*] Nmap: Discovered open port 80/tcp on 192.168.1.3
[*] Nmap: Discovered open port 22/tcp on 192.168.1.3
[*] Nmap: Discovered open port 139/tcp on 192.168.1.3
[*] Nmap: Discovered open port 445/tcp on 192.168.1.3
[*] Nmap: Discovered open port 23/tcp on 192.168.1.3
[*] Nmap: Discovered open port 53/tcp on 192.168.1.3
[*] Nmap: Discovered open port 21/tcp on 192.168.1.3
[*] Nmap: Discovered open port 3306/tcp on 192.168.1.3
[*] Nmap: Discovered open port 25/tcp on 192.168.1.3
[*] Nmap: Discovered open port 2121/tcp on 192.168.1.3
[*] Nmap: Discovered open port 6667/tcp on 192.168.1.3
[*] Nmap: Discovered open port 5432/tcp on 192.168.1.3
[*] Nmap: Discovered open port 1524/tcp on 192.168.1.3
[*] Nmap: Discovered open port 514/tcp on 192.168.1.3
[*] Nmap: Discovered open port 512/tcp on 192.168.1.3
[*] Nmap: Discovered open port 8180/tcp on 192.168.1.3
[*] Nmap: Discovered open port 513/tcp on 192.168.1.3
[*] Nmap: Discovered open port 8009/tcp on 192.168.1.3
[*] Nmap: Discovered open port 2049/tcp on 192.168.1.3
[*] Nmap: Discovered open port 6000/tcp on 192.168.1.3
[*] Nmap: Discovered open port 1099/tcp on 192.168.1.3
[*] Nmap: Completed SYN Stealth Scan at 22:00, 0.01s elapsed (1000 total ports)
[*] Nmap: Nmap scan report for 192.168.1.3
[*] Nmap: Host is up (0.00022s latency).
[*] Nmap: Not shown: 977 closed ports
[*] Nmap: PORT      STATE SERVICE
[*] Nmap: 21/tcp    open  ftp
```

```
[*] Nmap: 22/tcp    open  ssh
[*] Nmap: 23/tcp    open  telnet
[*] Nmap: 25/tcp    open  smtp
[*] Nmap: 53/tcp    open  domain
[*] Nmap: 80/tcp    open  http
[*] Nmap: 111/tcp   open  rpcbind
[*] Nmap: 139/tcp   open  netbios-ssn
[*] Nmap: 445/tcp   open  microsoft-ssn
[*] Nmap: 512/tcp   open  exec
[*] Nmap: 513/tcp   open  login
[*] Nmap: 514/tcp   open  shell
[*] Nmap: 1099/tcp  open  rmiregistry
[*] Nmap: 1524/tcp  open  ingreslock
[*] Nmap: 2049/tcp  open  nfs
[*] Nmap: 2121/tcp  open  ccproxy-ftp
[*] Nmap: 3306/tcp  open  mysql
[*] Nmap: 5432/tcp  open  postgresql
[*] Nmap: 5900/tcp  open  vnc
[*] Nmap: 6000/tcp  open  X11
[*] Nmap: 6667/tcp  open  irc
[*] Nmap: 8009/tcp  open  ajp13
[*] Nmap: 8180/tcp  open  unknown
[*] Nmap: MAC Address: BC:24:11:45:6B:0E (Unknown)
[*] Nmap: Read data files from: /usr/bin/./share/nmap
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
[*] Nmap: Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (40.120KB)
msf6 >
```

Phase 2: Using Metasploit port scanning tools

- Cropped screenshot of “search” and “use” commands to load and investigate the portscan syn package.

```
msf6 > search portscan

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/scanner/http/wordpress_pingback_access  normal  No  Wordpress Pingback Locator
1  auxiliary/scanner/natpmp/natpmp_portscan          normal  No  NAT-PMP External Port Scanner
2  auxiliary/scanner/portscan/ack                    normal  No  TCP ACK Firewall Scanner
3  auxiliary/scanner/portscan/ftpbounce              normal  No  FTP Bounce Port Scanner
4  auxiliary/scanner/portscan/syn                    normal  No  TCP SYN Port Scanner
5  auxiliary/scanner/portscan/tcp                    normal  No  TCP Port Scanner
6  auxiliary/scanner/portscan/xmas                    normal  No  TCP "XMas" Port Scanner
7  auxiliary/scanner/sap/sap_router_portscanner       normal  No  SAPRouter Port Scanner

Interact with a module by name or index. For example info 7, use 7 or use auxiliary/scanner/sap/sap_router_portscanner
```

```
msf6 > use auxiliary/scanner/portscan/syn
msf6 auxiliary(scanner/portscan/syn) > show
```

- META VM SYN Portscan:
 - Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.

```
msf6 auxiliary(scanner/portscan/syn) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/portscan/syn) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/portscan/syn) > show options

Module options (auxiliary/scanner/portscan/syn):
```

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	1-10000	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS	192.168.1.3	yes	The target host(s), range CIDR identifier, or hosts file with syntax 'file:<path>'
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	50	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

```
msf6 auxiliary(scanner/portscan/syn) >
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/portscan/syn) > run

[+] TCP OPEN 192.168.1.3:21
[+] TCP OPEN 192.168.1.3:22
[+] TCP OPEN 192.168.1.3:23
[+] TCP OPEN 192.168.1.3:25
[+] TCP OPEN 192.168.1.3:53
[+] TCP OPEN 192.168.1.3:80
[+] TCP OPEN 192.168.1.3:111
[+] TCP OPEN 192.168.1.3:139
[+] TCP OPEN 192.168.1.3:445
[+] TCP OPEN 192.168.1.3:512
[+] TCP OPEN 192.168.1.3:513
[+] TCP OPEN 192.168.1.3:514
[+] TCP OPEN 192.168.1.3:1099
[+] TCP OPEN 192.168.1.3:1524
[+] TCP OPEN 192.168.1.3:2049
[+] TCP OPEN 192.168.1.3:2121
[+] TCP OPEN 192.168.1.3:3306
[+] TCP OPEN 192.168.1.3:3632
[+] TCP OPEN 192.168.1.3:5432
[+] TCP OPEN 192.168.1.3:5900
[+] TCP OPEN 192.168.1.3:6000
[+] TCP OPEN 192.168.1.3:6667
[+] TCP OPEN 192.168.1.3:6697
[+] TCP OPEN 192.168.1.3:8009
[+] TCP OPEN 192.168.1.3:8180
[+] TCP OPEN 192.168.1.3:8787
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/portscan/syn) >
```

- Explanation of results from the scan output
 TCP SYN's were delivered to 192.168.1.3 by the SYN scan you performed (msf auxiliary scanner/portscan/syn), which also indicated which TCP ports were open. The module discovered numerous open ports i.e 21, 22, 23, 25, 53, 80, 111, 139, 445, 3306, 5432, 5900, 6000, 6667, 8009, 8180, 8787 with RHOSTS=192.168.1.3, PORTS=1-10000, and THREADS=50. To determine whether a host will accept TCP connections without completing the entire handshake, a SYN scan is a half-open (stealthy) scan.

Every open port, such as 21=FTP, 22=SSH, 23=Telnet, 25=SMTP, 53=DNS, 80=HTTP, 3306=MySQL, 5432=Postgres, 5900=VNC, 6000=X11, 6667=IRC, and 8009/AJP=app server connectors, is an attack surface. If you are permitted, the next steps are: Prioritize remediation, execute service/version detection (such as nmap -sV or Metasploit service modules), and look for known configuration errors or vulnerabilities. Stop testing this host if you are not specifically authorized to do so. Scanning can be invasive and should only be done within your permitted scope.

Part D: Intelligence gathering from targeted scanning of servers from KALI VM

Phase 1: Investigating SMB servers

- Target: META VM, Scanner: smb_version
- Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 auxiliary(scanner/portscan/syn) > search smb_version

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/scanner/smb/smb_version          normal         No     SMB Version Detection

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/smb/smb_version
msf6 auxiliary(scanner/portscan/syn) >

msf6 auxiliary(scanner/portscan/syn) > use auxiliary/scanner/smb/smb_version
msf6 auxiliary(scanner/smb/smb_version) >
```

- Cropped screenshot of “show options” after configuring RHOSTS for your scan.

```
msf6 auxiliary(scanner/smb/smb_version) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/smb/smb_version) > show options

Module options (auxiliary/scanner/smb/smb_version):

Name      Current Setting  Required  Description
-  -  -  -
RHOSTS    192.168.1.3     yes       The target host(s), range CIDR identifier, or hosts file with syntax 'file:filepath'
THREADS    1               yes       The number of concurrent threads (max one per host)

msf6 auxiliary(scanner/smb/smb_version) >
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/smb/smb_version) > run

[*] 192.168.1.3:445 - SMB Detected (versions:1) (preferred dialect:) (signatures:optional)
[*] 192.168.1.3:445 - Host could not be identified: Unix (Samba 3.0.20-Debian)
[*] 192.168.1.3: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_version) >
```

- Explanation of results from the scan output

The service on 192.168.1.3:445 was identified by the SMB scan as Samba 3.0.20 (Debian). Since that is an outdated version of Samba with known vulnerabilities, it is a high-priority lead for additional investigation, but only with the appropriate authorization. The immediate recommendations are to patch or upgrade Samba, examine shares and access controls, and (if permitted) perform secure, version-specific checks or vulnerability scans.

Phase 2: Investigating POSTGRES servers

- Target: META VM, Scanner: postgres_version
- Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 auxiliary(scanner/ssh/ssh_version) > back
msf6 > search postgres_version

Matching Modules

# Name Disclosure Date Rank Check Description
- -
0 auxiliary/scanner/postgres/postgres_version normal No PostgreSQL Version Probe

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/postgres/postgres_version
msf6 > use auxiliary/scanner/postgres/postgres_version
msf6 auxiliary(scanner/postgres/postgres_version) > 
```

- Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.

```
msf6 auxiliary(scanner/postgres/postgres_version) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/postgres/postgres_version) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/postgres/postgres_version) > show options

Module options (auxiliary/scanner/postgres/postgres_version):

Name Current Setting Required Description
- - - -
DATABASE template1 yes The database to authenticate against
PASSWORD postgres no The password for the specified username. Leave blank for a random password.
RHOSTS 192.168.1.3 yes The target host(s), range CIDR identifier, or hosts file with syntax 'file:<path>'
RPORT 5432 yes The target port
THREADS 50 yes The number of concurrent threads (max one per host)
USERNAME postgres yes The username to authenticate as
VERBOSE false no Enable verbose output

msf6 auxiliary(scanner/postgres/postgres_version) > 
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/postgres/postgres_version) > run

[*] 192.168.1.3:5432 Postgres - Version PostgreSQL 8.3.1 on i486-pc-linux-gnu, compiled by GCC cc (GCC) 4.2.3 (Ubuntu 4.2.3-2ubuntu4) (Post-Auth)
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/postgres/postgres_version) > 
```

- Explanation of results from the scan output
The database service on 192.168.1.3:5432 was successfully linked to by the PostgreSQL version scan, which revealed that it was PostgreSQL 8.3.1 operating on Linux (Ubuntu). This data demonstrates that the PostgreSQL service is operational and reachable with the standard login credentials (postgres/postgres). Version 8.3.1 is a possible security risk because it is extremely out-of-date and might have multiple known vulnerabilities. This result helps discover prospective attack pathways for privilege escalation or data access if correct security updates and configurations aren't in place.

Phase 3: Investigating SSH servers

- Target: META VM, Scanner: ssh_version
- Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 auxiliary(scanner/postgres/postgres_version) > back
msf6 > search ssh_version

Matching Modules

# Name Disclosure Date Rank Check Description
- -
0 auxiliary/fuzzers/ssh/ssh_version_15 normal No SSH 1.5 Version Fuzzer
1 auxiliary/fuzzers/ssh/ssh_version_2 normal No SSH 2.0 Version Fuzzer
2 auxiliary/fuzzers/ssh/ssh_version_corrupt normal No SSH Version Corruption
3 auxiliary/scanner/ssh/ssh_version normal No SSH Version Scanner

Interact with a module by name or index. For example info 3, use 3 or use auxiliary/scanner/ssh/ssh_version
msf6 > use auxiliary/scanner/ssh/ssh_version
msf6 auxiliary(scanner/ssh/ssh_version) > 
```

- Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.

```
msf6 auxiliary(scanner/ssh/ssh_version) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/ssh/ssh_version) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/ssh/ssh_version) > show options

Module options (auxiliary/scanner/ssh/ssh_version):



| Name    | Current Setting | Required | Description                                                                        |
|---------|-----------------|----------|------------------------------------------------------------------------------------|
| RHOSTS  | 192.168.1.3     | yes      | The target host(s), range CIDR identifier, or hosts file with syntax 'file:<path>' |
| RPORT   | 22              | yes      | The target port (TCP)                                                              |
| THREADS | 50              | yes      | The number of concurrent threads (max one per host)                                |
| TIMEOUT | 30              | yes      | Timeout for the SSH probe                                                          |



msf6 auxiliary(scanner/ssh/ssh_version) > █
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/ssh/ssh_version) > run

[*] 192.168.1.3:22 - SSH server version: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1 ( service.version=4.7p1 openssh.com
ment-Debian-8ubuntu1 service.vendor=OpenBSD service.family=OpenSSH service.product=OpenSSH service.cpe23=cpe:/a:openbsd:ope
nssh:4.7p1 os.vendor=Ubuntu os.family=Linux os.product=Linux os.version=8.04 os.cpe23=cpe:/o:canonical:ubuntu_linux:8.0
4 service.protocol=ssh fingerprint_db=ssh.banner )
[*] 192.168.1.3:22 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ssh/ssh_version) > █
```

- Explanation of results from the scan output

The target system at 192.168.1.3 is running a OpenSSH 4.7p1 server on Debian 8 (Ubuntu 8.04), according to the results of the SSH version scan. This output indicates the underlying operating system and software version and verifies that port 22 (SSH) is open. Both Ubuntu 8.04 and OpenSSH 4.7p1 are out-of-date and may have known security flaws that, if left unpatched, could be exploited. Although exploitation efforts should only be carried out with express permission, this information is useful for evaluating possible attack surfaces.

Phase 4: Investigating FTP servers

- Target: META VM, Scanner: ftp_version
- Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 > search ftp_version

Matching Modules



| # | Name                              | Disclosure Date | Rank   | Check | Description         |
|---|-----------------------------------|-----------------|--------|-------|---------------------|
| 0 | auxiliary/scanner/ftp/ftp_version |                 | normal | No    | FTP Version Scanner |



Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/ftp/ftp_version

msf6 > use auxiliary/scanner/ftp/ftp_version
msf6 auxiliary(scanner/ftp/ftp_version) > █
```

- Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.

```
msf6 auxiliary(scanner/ftp/ftp_version) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/ftp/ftp_version) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/ftp/ftp_version) > show options

Module options (auxiliary/scanner/ftp/ftp_version):



| Name    | Current Setting     | Required | Description                                                                         |
|---------|---------------------|----------|-------------------------------------------------------------------------------------|
| FTPPASS | mozilla@example.com | no       | The password for the specified username                                             |
| FTPUSER | anonymous           | no       | The username to authenticate as                                                     |
| RHOSTS  | 192.168.1.3         | yes      | The target host(s), range CIDR identifier, or hosts file with syntax 'file:<pa th>' |
| RPORT   | 21                  | yes      | The target port (TCP)                                                               |
| THREADS | 50                  | yes      | The number of concurrent threads (max one per host)                                 |



msf6 auxiliary(scanner/ftp/ftp_version) > █
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/ftp/ftp_version) > run
[+] 192.168.1.3:21      - FTP Banner: '220 (vsFTPd 2.3.4)\x0d\x0a'
[*] 192.168.1.3:21      - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ftp/ftp_version) > █
```

- Explanation of results from the scan output
An FTP service operating on port 21 of the target host, 192.168.1.3, was detected by the FTP version scan, and its banner was revealed to be vsFTPd 2.3.4. This verifies that the FTP service is operational and reachable, and depending on the setup, it even permits anonymous login attempts. Because it has traditionally been linked to known vulnerabilities, such as a backdoor exploit in some versions, the version number (2.3.4) is noteworthy. This data aids in identifying any potential outdated or incorrectly configured FTP servers, which could present a security risk if not adequately patched or secured.
- Response to question: Cropped screenshot and explanation of the difference of outputs after running the “services” command again compared to the outputs obtained earlier.
While the second output, which is generated after performing targeted version/banner scans, contains richer information (software names, versions, and OS hints like vsFTPd 2.3.4, OpenSSH 4.7p1, Samba 3.0.20, and PostgreSQL 8.3.1), the first services output only provides a list of open ports and their common service names (basic port discovery). To put it briefly, the first output indicates what is open, and the second output indicates what is actually operating (helpful for identifying known vulnerabilities).

```
msf6 > services
Services
```

host	port	proto	name	state	info
192.168.1.3	21	tcp	ftp	open	220 (vsFTPd 2.3.4)\x0d\x0a
192.168.1.3	22	tcp	ssh	open	SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
192.168.1.3	23	tcp	telnet	open	
192.168.1.3	25	tcp	smtp	open	
192.168.1.3	53	tcp	domain	open	
192.168.1.3	80	tcp	http	open	
192.168.1.3	111	tcp	rpcbind	open	
192.168.1.3	139	tcp	netbios-ssn	open	
192.168.1.3	445	tcp	smb	open	Unix (Samba 3.0.20-Debian)
192.168.1.3	512	tcp	exec	open	
192.168.1.3	513	tcp	login	open	
192.168.1.3	514	tcp	shell	open	
192.168.1.3	1099	tcp	rmiregistry	open	
192.168.1.3	1524	tcp	ingreslock	open	
192.168.1.3	2049	tcp	nfs	open	
192.168.1.3	2121	tcp	ccproxy-ftp	open	
192.168.1.3	3306	tcp	mysql	open	
192.168.1.3	3632	tcp		open	
192.168.1.3	5432	tcp	postgres	open	PostgreSQL 8.3.1 on i486-pc-linux-gnu, compiled by GCC cc (GCC) 4.2.3 (Ubuntu 4.2.3-2ubuntu4)
192.168.1.3	5900	tcp	vnc	open	
192.168.1.3	6000	tcp	x11	open	
192.168.1.3	6667	tcp	irc	open	
192.168.1.3	6697	tcp		open	
192.168.1.3	8009	tcp	ajp13	open	
192.168.1.3	8180	tcp	unknown	open	
192.168.1.3	8787	tcp		open	

Phase 5: Further investigating FTP servers

- Target: META VM, Scanner: anonymous
 - Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 > grep scanner grep ftp search anonymous
11 auxiliary/scanner/ftp/anonymous normal No Anonymous FTP Access Detection
msf6 >
msf6 > use auxiliary/scanner/ftp/anonymous
msf6 auxiliary(scanner/ftp/anonymous) >
```

- Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.

```
msf6 auxiliary(scanner/ftp/anonymous) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/ftp/anonymous) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/ftp/anonymous) > show options
Module options (auxiliary/scanner/ftp/anonymous):


| Name    | Current Setting     | Required | Description                                                                       |
|---------|---------------------|----------|-----------------------------------------------------------------------------------|
| FTPPASS | mozilla@example.com | no       | The password for the specified username                                           |
| FTPUSER | anonymous           | no       | The username to authenticate as                                                   |
| RHOSTS  | 192.168.1.3         | yes      | The target host(s), range CIDR identifier, or hosts file with syntax 'file:paths' |
| RPORT   | 21                  | yes      | The target port (TCP)                                                             |
| THREADS | 50                  | yes      | The number of concurrent threads (max one per host)                               |


msf6 auxiliary(scanner/ftp/anonymous) >
```

- Cropped screenshot of output after running the scan

```
msf6 auxiliary(scanner/ftp/anonymous) > run
[+] 192.168.1.3:21 - 192.168.1.3:21 - Anonymous READ (220 (vsFTPD 2.3.4))
[*] 192.168.1.3:21 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ftp/anonymous) >
```

- Explanation of results from the scan output
According to the results of the FTP anonymous scan, the target host, 192.168.1.3, permits anonymous FTP access on port 21 through the vsFTPD 2.3.4 service. If important files are available, this presents a serious security concern because anyone can access the FTP server without authentication. Unauthorized people may be able to see or download files from the system due to a server that is misconfigured or insecurely set up, as the scan revealed successful anonymous read access.

Phase 6: Performing SNMP sweeping

- Target: META VM, Scanner: snmp_login
- Cropped screenshot of search result for locating scanner in msfconsole

```
msf6 auxiliary(scanner/ftp/anonymous) > back
msf6 > search snmp_login
Matching Modules


| # | Name                              | Disclosure Date | Rank   | Check | Description                  |
|---|-----------------------------------|-----------------|--------|-------|------------------------------|
| 0 | auxiliary/scanner/snmp/snmp_login |                 | normal | No    | SNMP Community Login Scanner |


Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/snmp/snmp_login
msf6 > use auxiliary/scanner/snmp/snmp_login
msf6 auxiliary(scanner/snmp/snmp_login) >
```

- Cropped screenshot of “show options” after configuring RHOSTS and THREADS for your scan.


```
msf6 auxiliary(scanner/snmp/snmp_login) > set RHOSTS 192.168.1.3
RHOSTS => 192.168.1.3
msf6 auxiliary(scanner/snmp/snmp_login) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/snmp/snmp_login) > show options

Module options (auxiliary/scanner/snmp/snmp_login):
```

Name	Current Setting	Required	Description
BLANK_PASSWORDS	false	no	Try blank passwords
BRUTEFORCE_SPEED	5	yes	How fast to bruteforce, from 0 to 5
DB_ALL_CREDS	false	no	Try each user/password couple stored in the current database
DB_ALL_PAS	false	no	Add all passwords in the current database to the list
DB_ALL_USERS	false	no	Add all users in the current database to the list
PASSWORD		no	The password to test
PASS_FILE	/usr/share/metasploit-framework/data/wordlists/snmp_default_pass.txt	no	File containing community strings, one per line
RHOSTS	192.168.1.3	yes	The target host(s), range CIDR identifier, or hosts file with syntax 'file: <path>'
RPORT	161	yes	The target port
STOP_ON_SUCCESS	false	yes	Stop guessing when a credential works for a host
THREADS	50	yes	The number of concurrent threads (max one per host)
USER_AS_PASS	false	no	Try the username as the password for all users
VERBOSE	true	yes	Whether to print output for all attempts
VERSION	1	yes	The SNMP version to scan (Accepted: 1, 2c, all)

```
msf6 auxiliary(scanner/snmp/snmp_login) >
```

- Cropped screenshot of output after running the scan successfully.

```
msf6 auxiliary(scanner/snmp/snmp_login) > run
[*] 192.168.1.3:161 - Login Successful: private (Access level: read-write); Proof (sysDescr.0): Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
[*] 192.168.1.3:161 - Login Successful: public (Access level: read-only); Proof (sysDescr.0): Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/snmp/snmp_login) >
```

- Explanation of results from the scan output
The target host 192.168.1.3 had two successful SNMP community string logins, according to the SNMP login scan: public with read only access and private with read write access. This indicates that both of the default community strings were active and the SNMP service port 161 was operating. This is a serious security concern since it allows hackers to view and alter system configurations. In order to profile potential vulnerabilities or exploits for that system, the scan output also confirmed the operating system version and architecture, identifying the system as Linux Metasploitable 2.6.24.